Can an Injected-Failure Benchmark Pick Your Monitor? A Four-Part Validity Audit for Enterprise Agent Deployment

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Abstract

An enterprise that runs agents in production needs to detect when one fails, and it usually picks the monitor that scores best on a benchmark of labelled failures. Those labels are cheap to obtain by injection: change a step of a successful trajectory and the benchmark knows which step it changed. A correct label, however, does not show that the injected failure stands in for the failures the monitor will meet in service. We treat the injector as part of the evaluation and ask four questions. Can a detector read data written only by the injector? When two methods create the same error, can it tell which method was used? On the same task, can it tell injected and natural failures apart? Would injected and natural data lead us to pick the same monitor? On 24,000 agent traces the four tests give different answers. Removing injector-only data erases some apparent detection ability. One injection method stays identifiable after we control measured error severity. Matching on the query removes an apparent realism trend, and the monitor chosen from injected data changes with the detector set. The gap matters at deployment: a monitor selected on injected traces can lose measurable accuracy under the shift to natural failures. The audit gives enterprise benchmark builders a practical way to state which claims their injected failures support before a monitor goes into production.

1 Why a clear label is not enough to pick a monitor

An agent running an enterprise workflow fails in ways nobody watches in real time, so teams attach an automatic monitor and choose it by how well it scores on a benchmark of labelled failures. Building that benchmark is the hard part: a failed trajectory records what happened but does not identify the step that caused the outcome. AEGIS, Who&When Pro, and AgenTracer therefore change successful trajectories and use the changed step as the label [Kong et al., 2026, Liu et al., 2026, Zhang et al., 2025]. This gives a clear label without finding the cause afterward.

The injected failure may still differ from a natural one. For example, a benchmark may know which supporting passage it removed, while the replacement rule leaves an obvious marker. A detector can then score well for the wrong reason. Prior work studies whether benchmarks measure their intended concept [Bean et al., 2025] and shows gaps between synthetic and natural data [CH-Wang et al., 2024, Chen et al., 2026, van Elburg et al., 2025]. We turn these concerns into four tests for injected agent benchmarks:

V1 Independence: Can the detector read data written only by the injector?

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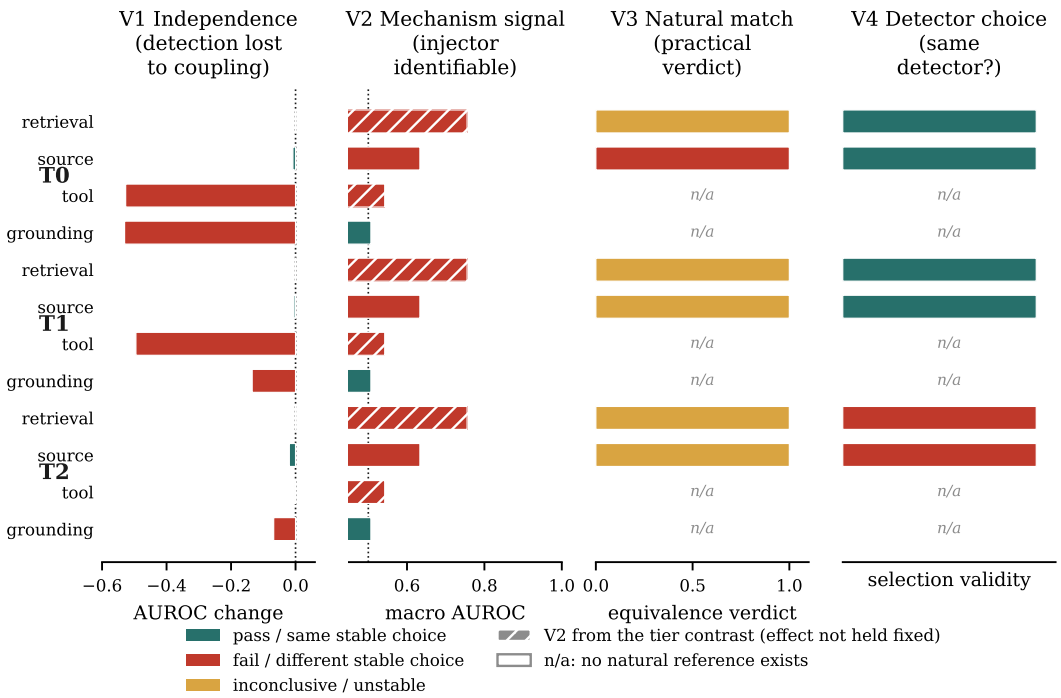


Figure 1: The four tests give different answers on the same benchmark conditions. Undefined and inconclusive cells are not counted as passes.

V2 Mechanism signal: When two methods create the same error, can the detector tell which method was used?

V3 Natural match: On the same task, can it tell injected and natural failures apart?

V4 Detector choice: Would injected and natural data lead us to choose the same detector, and how much natural-data performance is lost if they do not?

2 Experimental setup

The agent pipeline uses BGE retrieval and reranking, Qwen2.5-7B-Instruct for generation and judging, and real tool execution. The main dataset contains 24,000 HotpotQA and GSM8K traces. We inject four kinds of failures involving the query, supporting sources, tool use, and final answer. Each has naive (T0), literature-inspired (T1), and artifact-hardened (T2) versions. Three of fourteen detector features read injector-written data directly; the main feature set removes them. Training and test folds never share a query. We use 1,000 query-level bootstrap samples for uncertainty.

HotpotQA labels which passages support an answer. We call a trace a naturally occurring retrieval degradation when the selected context omits a required passage. A separate 2Wiki run adds 7,000 traces, including 521 natural degradations and 6,000 injected traces on the same queries.

3 Results

V1: injector-only data can inflate scores. Tool-fault detection at T1 falls from .990 to .495 after removing the three direct links; tool T0 falls from 1.000 to .473, and grounding T0 from 1.000 to .469. Retrieval T1 (.864) and source T1 (.963) change little. The losses are therefore tied to the direct links rather than a general loss of signal.

V2: the injection method can leave a signal. Two source injectors replace one supporting passage in the same slot with a rejected passage from the same query. A classifier identifies which method was used with AUROC .633 [.601, .664]. Measured error severity explains part of this result. After

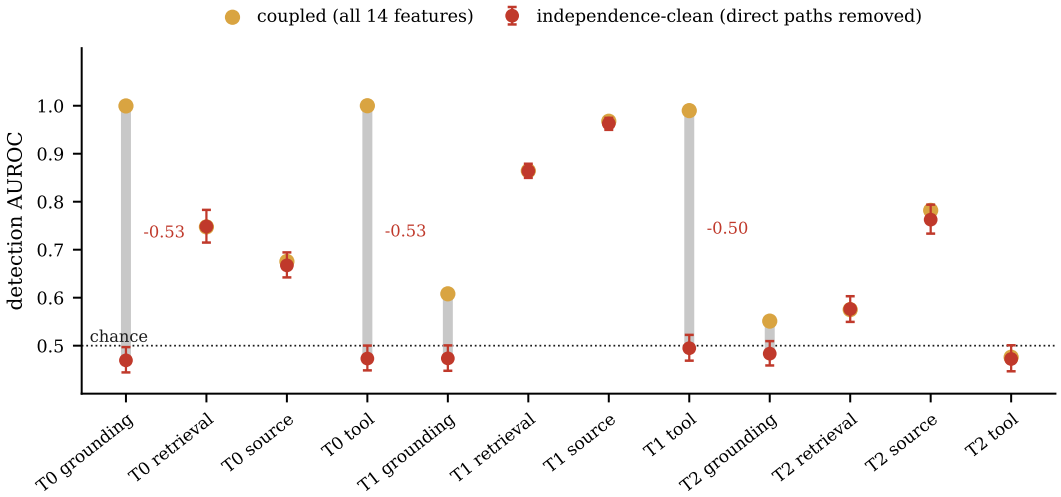


Figure 2: Detection AUROC before and after removing features that directly read injector data. Large losses occur only in conditions with a direct link.

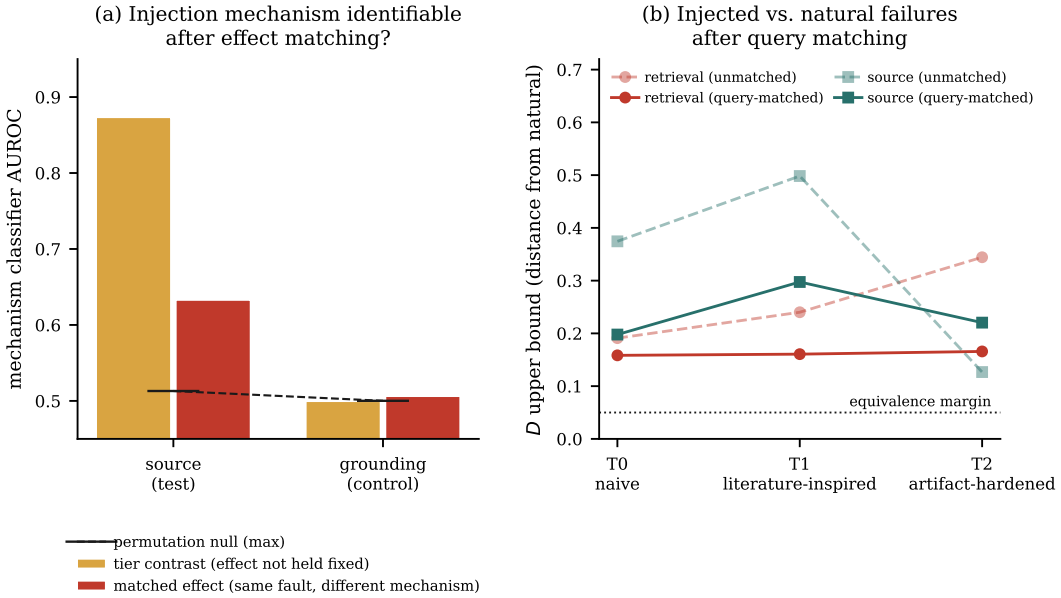


Figure 3: (a) Can a classifier identify the injection method after the intended error is matched? (b) Can it tell injected and natural failures apart on the same queries?

reweighting eight severity measures, the classifier still reaches .607 [.576, .639]. A similar grounding comparison is at chance (.504). The signal therefore appears for one tested fault type but not the other.

V3: the query must be matched. Without matching, source faults appear closer to natural failures as the injector is hardened, while retrieval faults move the other way. Natural failures, however, occur on queries the retriever finds hard. The pattern disappears when injected and natural traces use the same queries. No Hotpot condition establishes a practical match at $\delta = .05$; most results are inconclusive rather than proven different. On 2Wiki, all six conditions are distinguishable, but hardening still has no steady direction.

V4: the detector set changes the answer. With all seven detectors, Hotpot T0/T1 choose the natural-data winner. T2 chooses a different detector but loses only .004 AUROC on natural data.

Table 1: Each benchmark claim needs different evidence. A failed or inconclusive test limits only the claim in its row.

Claim	Required evidence	Evidence here
Score does not use injector-only data	V1 after direct links are removed	passes for several retrieval/source cells; fails for three coupled cells
Injection method does not matter	V2 after matching the error and severity	source signal remains; grounding is at chance
Injected failures match natural ones	V3 on the same queries	no practical match at $\delta = .05$ on either dataset
Injected data choose the natural-data detector	V4 choice stability and natural-data loss	answer changes with dataset and detector set

After removing detectors that fail V1, T0/T1 lose .058, while T2 chooses the same detector in .990 of bootstrap samples. On 2Wiki, the V1-clean T0/T2 choices lose .088 [.059, .115]. Detector choice therefore depends on the benchmark, natural data, metric, and detectors being compared.

4 Checking the natural comparison

We add the missing gold passage back to wrong-answer traces and rerun the agent. On HotpotQA, .311 of answers become correct, compared with .100 when we add a different non-gold passage of similar length. On 2Wiki, the rates are .332 and .107. Missing evidence therefore contributes to some wrong answers, but it is not a complete cause label for every case. On 600 released AEGIS traces, removing available injector fields changes AUROC by .032 [.001, .066] and .022 [-.011, .058] for two modes. Both intervals cross the .05 margin, so these external V1 results are inconclusive.

5 A protocol for enterprise benchmark builders

Before using injected failures to compare detectors: (1) remove direct paths from injector data to the detector; (2) build two methods that create the same error and test whether the method can be identified; (3) compare injected and natural failures on the same tasks; (4) check whether both sources choose the same detector and report the natural-data loss when they do not; and (5) report undefined or inconclusive tests instead of treating them as passes. For an enterprise deployment, step (4) carries the cost: a monitor chosen on injected traces can lose measurable accuracy under the shift to the natural failures it meets in service.

Limits and conclusion. The study uses one agent pipeline and one generator. Natural comparisons exist only for retrieval-side faults, and V2 matches the intended error for only two of four fault types. Failure to identify a difference under the tested features does not prove that the full trace distributions are equal. The appendix gives the complete settings, features, injector operations, uncertainty results, and second-dataset tables. **Clear injected labels are useful, but a benchmark that has not been audited cannot tell an enterprise which monitor to trust in production.**

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A Operating-point and second-dataset details

HotpotQA k=3. Because k=2 was chosen after inspecting the natural-failure rate, we repeat the natural and decision analyses at k=3. Natural failures fall to 17.25% $(138/800)$, while wrongness lift remains positive at $+0.292$ [+0.205, +0.374]. Under Set A, T0/T1 agree and T2 misselects (regret 0.015, bootstrap agreement $.038$); under Set B, T0/T1 misselect while T2 agrees only unstably ($P_{\text{agree}} = .573$). The qualitative candidate-pool interaction survives the operating point, but its individual point winners need not be stable.

2Wiki protocol. A 200-query validation pilot, excluded from all later analysis, observed 114 natural degradations and projected 570 at the fixed cap of 1,000. The confirmatory partition contains exactly 1,000 clean and 6,000 synthetic traces: T0–T2 crossed with retrieval and source degradation. Query-disjoint five-fold predictions, within-query permutation nulls and 1,000 query-cluster bootstraps are unchanged from the primary protocol. Both partitions use the `framolfese/2WikiMultihopQA` validation split with seed `20260823`. Among the 521 confirmatory natural degradations, 302 omit one annotated supporting title and 219 omit multiple (213 omit two and 6 omit three); their wrong-answer rates are .517, .141, and .167, respectively. We therefore retain the missing-evidence count as a severity descriptor rather than interpreting the pooled natural class as severity-homogeneous.

Table 2: Full 2Wiki decision-validity replication. Regret is measured on natural failures; intervals and P_{agree} use query-cluster bootstrap resamples.

Set	Tier	Synthetic choice	Natural choice	P_{agree}	Regret [95% CI]
All	T0	all features	all features	.963	.000 [.000, .100]
All	T1	all features	all features	.692	.000 [.000, .109]
All	T2	all features	all features	.755	.000 [.000, .117]
V1-clean	T0	semantic	decoupled	.000	.088 [.059, .115]
V1-clean	T1	decoupled	decoupled	.816	.000 [.000, .101]
V1-clean	T2	semantic	decoupled	.000	.088 [.060, .115]

Table 3: All 2Wiki detector AUROCs. Natural scores detect missing annotated evidence on clean traces; synthetic columns pool the two comparable fault families within each tier.

Detector	Natural	T0	T1	T2
all features	.839	.795	.933	.787
decoupled	.823	.773	.932	.782
semantic only	.735	.794	.932	.787
trivial baselines	.752	.532	.608	.557
NLI	.605	.539	.521	.524
NLI-LARGE	.631	.510	.580	.543
JUDGE	.671	.575	.695	.615

Table 4: 2Wiki V3 equivalence sensitivity under the held-out external representation. A pass requires the upper confidence bound on D to fall strictly below the displayed margin.

Condition	δ=.05	δ=.10	δ=.15	δ=.20
T0 retrieval	fail	fail	fail	fail
T1 retrieval	fail	fail	fail	fail
T2 retrieval	fail	fail	pass	pass
T0 source	fail	fail	pass	pass
T1 source	fail	fail	fail	fail
T2 source	fail	fail	fail	fail

B External AEGIS audit protocol

We pin AEGIS at commit `18fecc5641235a67299cd832632839c9c80244d6` and retain all 600 released rows. FM-1.1 and FM-2.2 were fixed before the confirmatory run. Five folds are task-disjoint; intervals use 1,000 task-cluster bootstraps, and raw out-of-fold predictions are stored in the result JSON. Splitting and resampling use seed `20260823`.

The final row has a direct injection-side path and is excluded from both primary arms. The available arm therefore tests the incremental value of ordinary benchmark metadata, while the decoupled arm is trajectory text alone. The released sample supports only 61 successful single-injection rows across frameworks containing both prompt and response strategies; this is why its V2 analysis is labelled diagnostic rather than effect-matched.

C Complete confirmatory statistics

Table 6 reports every primary held-out-representation V3 cell rather than only the upper bounds summarized in the main paper. HotpotQA contributes five injected traces per natural-failure query

Table 5: Field-level AEGIS causal-path audit. The unsafe arm is a negative control and never supports a primary result.

Fields	Available	Decoupled	Unsafe diagnostic
query, conversation history, final output	yes	yes	yes
framework, benchmark, model, agent count, task type	yes	no	yes
`output.faulty_agents`, injected-agent targets, injection-success flag, injected-agent count	no	no	yes

because the primary campaign has five injector seeds; 2Wiki contributes one. The unit of resampling and splitting remains the query in both cases.

Table 6: Complete query-matched V3 results. AUROC intervals are query-cluster bootstrap intervals; D_{upper} is the one-sided practical-equivalence bound. "Inc." means that chance remains inside the interval but equivalence at $\delta = .05$ is not established.

Dataset	Condition	Queries	AUROC [95% CI]	D	D_{upper}	Verdict
HotpotQA	T0 retrieval	71	.508 [.437, .579]	.016	.158	inc.
	T1 retrieval	71	.486 [.420, .556]	.029	.161	inc.
	T2 retrieval	71	.460 [.417, .504]	.080	.166	inc.
	T0 source	71	.558 [.528, .599]	.117	.198	dist.
	T1 source	71	.552 [.457, .649]	.105	.297	inc.
	T2 source	71	.459 [.390, .518]	.082	.220	inc.
2Wiki	T0 retrieval	521	.607 [.577, .639]	.215	.278	dist.
	T1 retrieval	521	.652 [.617, .684]	.303	.369	dist.
	T2 retrieval	521	.539 [.510, .569]	.078	.138	dist.
	T0 source	521	.556 [.542, .570]	.112	.141	dist.
	T1 source	521	.599 [.570, .629]	.197	.257	dist.
	T2 source	521	.590 [.565, .615]	.179	.230	dist.

Table 7: Complete AEGIS V1 results. The difference is available minus decoupled AUROC; a material coupling loss is declared only when its interval lies above .05.

Mode	Pos./total	Available	Decoupled	Difference [95% CI]	Verdict
FM-1.1	139/600	.673 [.612, .727]	.641 [.579, .695]	.032 [.001, .066]	inc.
FM-2.2	98/600	.631 [.571, .694]	.609 [.545, .672]	.022 [−.011, .058]	inc.

The AEGIS diagnostic V2 subset contains 26 prompt-side and 35 response-side successful single injections across `dylan` and `llm_debate`. Strategy AUROC is .407 [.238, .569]; reversing the arbitrary binary orientation would give .593. Its width and unmatched effects preclude either a pass or a failure verdict.

D Representations, estimators and missing values

The primary representation consists of the fourteen quantities in Table 9. Every cosine uses the normalized BGE embedding. A missing metric is never replaced by an extreme sentinel: within each training fold we median-impute it and add a separate definedness indicator, then apply that fitted transform unchanged to the held-out fold. This prevents inapplicability from masquerading as an extreme metric value while still allowing the probe to use the fact that a quantity is undefined.

The primary probe is a class-balanced logistic regression with standardization and `max_iter=5000`. All reported predictions are out of fold under five query-disjoint folds. The nonlinear sensitivity analysis changes only the estimator: an RBF-SVM with balanced classes, default gradient boosting, and a standardized $(64, 32)$ MLP trained for at most 800 iterations, all untuned and using the same folds.

V3 deliberately uses representations not consulted while designing T2. The surface arm has nine raw-text statistics: answer characters and tokens, passage count, context characters, mean passage length, answer–context and question–context token overlap, answer type-token ratio and maximum answer-token frequency. The external arm contains cached scores from `potsawee/deberta-v3-large-mnli`,

Table 8: Complete causal validation of the natural comparator. Discordances are gold-only/placebo-only rescues; p is the exact McNemar test.

Dataset	Intervened	Verified	Gold rescue	Placebo rescue	Effect [95% CI]	Discord.; p
HotpotQA	90	19	.311 [.211, .411]	.100 [.044, .156]	.211 [.122, .300]	19/0; 3.81×10^{-6}
2Wiki	187	50	.332 [.262, .401]	.107 [.064, .150]	.225 [.150, .294]	50/8; 1.57×10^{-8}

Table 9: Complete feature dictionary. q, a, c_i, r_i, g_i, s_i denote the question, answer, selected passages, rejected passages, subgoals and step-context centroids. A dagger marks a feature removed in the V1-decoupled arm.

Feature	Operational definition
QCM	mean $\cos(q, c_i)$ minus mean $\cos(q, r_i)$
ICRR	mean pairwise cosine among selected passages
SCR	$1 - \max_i \cos(q, c_i)$
$\text{CDI}_{\max}$	largest change in SCR after deleting one selected passage
MUT	$\max_i \cos(q, c_i)$ divided by selected-context token count
IPS	mean $\cos(q, g_i)$ over generated subgoals
TCS	mean pairwise cosine among step-context centroids
GSA	mean $\cos(q, s_i)$ over step-context centroids
SRS	mean over steps of $\max_i \cos(s_j, c_i)$
AGS†	mean $\cos(a, c_i)$ over selected passages
$\text{TRTR}^\dagger_{\text{route}}$	fraction of tool calls whose name is not registered
$\text{TIHD}^\dagger_{\text{health}}$	fraction of tool calls that return a failed status
token count	whitespace-token count of selected context
answer length	whitespace-token count of the generated answer

`MoritzLaurer/DeBERTa-v3-large-mnli-fever-anli-ling-wanli`, and a frozen Qwen2.5-7B-Instruct JSON judge. The first NLI score applies SelfCheckGPT-NLI's contradiction scoring rule to evidence–answer pairs; it is not the full SelfCheckGPT method.

AEGIS uses a separate text representation because its released schema is not a MIRAGE trace. Each arm is vectorized with TF–IDF word uni/bi-grams (`min_df=2`, `max_features=20000`, sublinear TF), followed by class-balanced logistic regression. No MIRAGE metric or injector label is transferred into this audit.

E Natural-data budget and mixing protocols

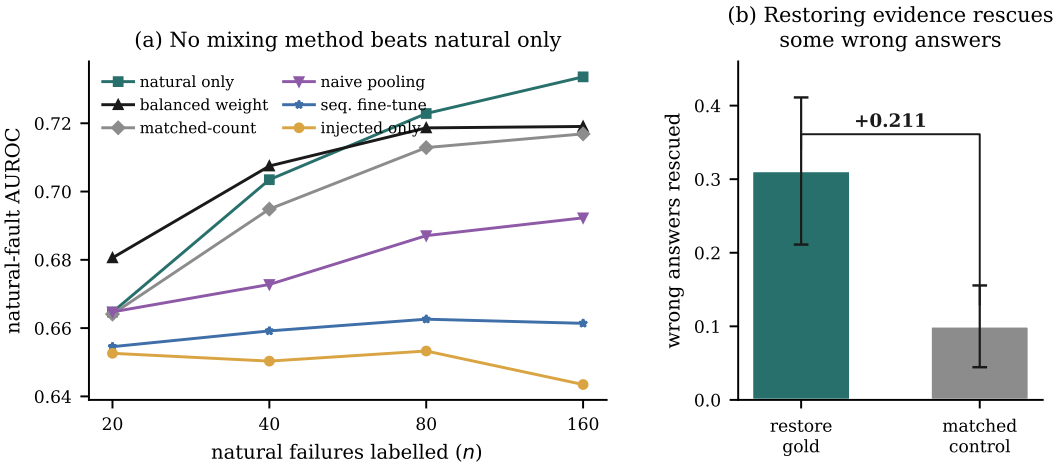


Figure 4: (a) Natural-fault AUROC against labelled natural failures, over 40 query-level draws. (b) Rescue rates under gold restoration and a length-matched placebo.

A naive comparison would be confounded: thousands of synthetic rows against twenty natural ones dominate any pooled objective by count alone. We compare six protocols on identical folds — synthetic only, natural only, naive pooling, balanced domain weighting, sequential fine-tuning and matched-count augmentation — and report paired differences against natural-only.

No protocol significantly outperforms natural-only at any budget. Naive pooling, balanced weighting and matched-count are statistically indistinguishable from natural-only throughout; balanced weighting is numerically ahead at n=20 ($+0.016$ [-0.042, +0.097]). Only synthetic-only and sequential fine-tuning become significantly worse, from n=80 (-0.070 [-0.105, -0.010] and -0.060 [-0.156, -0.000]). At n=160 synthetic-only trails by -0.090 [-0.124, -0.035]. This does not support the broader claim that twenty real labels beat thousands of synthetic ones; the gap at n=20 is not significant. We make no claim about annotation cost, which we did not measure.

F Detection against external detectors

Table 10: Detection AUROC on the DECOUPLED arm under identical query-disjoint folds. Best per row in bold; the all-features column indicates the size of the coupling correction.

Tier	Family	DECOUPLED	NLI-LARGE	NLI	JUDGE	TRIVIAL	*(all feat.)*
T0	retrieval	**0.748**	0.497	0.475	0.635	0.698	*0.748*
T0	source	**0.667**	0.522	0.564	0.538	0.514	*0.675*
T0	tool	0.473	0.527	0.478	0.489	**1.000**	*1.000*
T0	grounding	0.469	0.795	0.570	**0.954**	0.476	*1.000*
T1	retrieval	**0.864**	0.489	0.501	0.652	0.699	*0.864*
T1	source	**0.963**	0.759	0.605	0.808	0.633	*0.968*
T1	tool	0.495	**0.536**	0.477	0.494	0.562	*0.990*
T1	grounding	0.474	0.748	0.558	**0.772**	0.516	*0.608*
T2	retrieval	**0.576**	0.516	0.475	0.526	0.528	*0.575*
T2	source	**0.763**	0.618	0.538	0.616	0.573	*0.782*
T2	tool	0.472	0.491	0.482	0.487	**0.516**	*0.476*
T2	grounding	0.483	**0.700**	0.557	0.691	0.520	*0.551*

Combining detectors does not help. Under a paired cluster bootstrap on pooled out-of-fold predictions, 10 of 36 union comparisons are significant: eight are regressions, all on grounding (largest -0.135 [-0.156, -0.113]), and two are small gains. Adding eleven features blind to grounding dilutes the one or two that are not, so the recommendation is to route by fault family rather than pool.

G The full injector ladder

Table 11: All twelve (tier, family) operations. Each injector declares how much of the pipeline must be re-executed downstream.

Family	Tier	Operation
`query_retrieval`	T0	Reverse the word order of the query.
	T1	Random dropout of $\approx 50\%$ of query tokens.
	T2	Delete one high-information content word, chosen by seeded RNG from the three longest.
`source_degradation`	T0	Prefix the passage with `"OUTDATED "`.
	T1	Exchange the selected and rejected passage sets.
	T2	Keep the top passage; replace the remainder with near-miss distractors.
`tool_invocation`	T0	Call an unregistered tool name.
	T1	Emit a malformed expression that really executes and errors.
	T2	Emit a valid expression with two operands transposed.
`grounding`	T0	Swap in a donor answer from another query.
	T1	Substitute an entity in the model's own answer.
	T2	Generator rewrites its answer so one RNG-chosen detail is unsupported.

Table 12: Matched-effect mechanism identification (DECOUPLED arm). Contrasts involving `donor` are reported for completeness; `donor` draws from an unrelated query and is topically off-subject, so it is not the basis of any claim.

source_degradation	rank–lexical	rank–donor	lexical–donor
	0.633	0.989	0.984
grounding	dict–distractor	dict–donor	distractor–donor
	0.507	0.507	0.499

Table 13: Declared severity balance for the defensible rank–lexical source contrast. Absolute standardised mean differences are shown before and after clipped cross-fitted propensity weighting.

Severity variable	Before	After
answer similarity	.029	.002
answer token-length change	.018	.002
answer character-length change	.011	.010
changed-token count	.016	.006
correctness change	.009	.001
answer token-F1 change	.041	.004
context token-length change	.320	.014
question–context overlap	.374	.002

H Matched-effect contrasts and tier contrasts

Severity-only AUROC is .628 [.604, .653]. After reweighting, the decoupled mechanism probe is .607 [.576, .639]; weights range from .603 to 1.966. The signal attenuates but remains beyond $D = .05$, while maximum imbalance falls from .374 to .014.

Table 14: Within-family pairwise *tier* discrimination (DECOUPLED). These do not hold the semantic effect fixed and are superseded above where a matched construction exists.

Family	T0–T1	T0–T2	T1–T2
source_degradation	0.957	0.768	0.949
query_retrieval	0.813	0.717	0.801
tool_invocation	0.586	0.542	0.541
grounding	0.499	0.512	0.500

I Probe-family sensitivity

Discrimination scale D under four model families on identical query-disjoint folds, untuned. No non-linear family exceeds the linear probe by more than $+0.038$, so the negative results in Sections 3 and 3 are not artifacts of linearity.

Condition	linear	RBF-SVM	GBM	MLP
T0 retrieval	0.454	0.241	0.373	0.324
T0 source	0.301	0.295	0.206	0.286
T1 retrieval	0.655	0.567	0.607	0.528
T1 source	0.708	0.554	0.536	0.626
T2 retrieval	0.224	0.142	0.005	0.128
T2 source	0.145	0.112	0.182	0.159

J Equivalence margin sensitivity

Query-matched C2ST under the external representation. Only at $\delta = 0.20$ do most conditions pass, so the conclusion in Section 3 is not an artifact of a strict margin: it holds across every margin we consider defensible.

Condition	δ=0.05	δ=0.10	δ=0.15	δ=0.20
T0 retrieval	fail	fail	fail	pass
T1 retrieval	fail	fail	fail	pass
T2 retrieval	fail	fail	fail	pass
T0 source	fail	fail	fail	pass
T1 source	fail	fail	fail	fail
T2 source	fail	fail	fail	fail

K Scale and reproducibility

The primary HotpotQA/GSM8K campaign contains 24,000 traces: E1 (200 clean), E3 (5,000; 200 queries, three seeds, probe training) and E4 (18,800; 800 disjoint queries, five seeds, test), generated in 15.2 A6000 GPU-hours. Additional versioned artifacts contain 7,000 2Wiki traces, 1,400 at HotpotQA k=3, 2,352 matched-effect traces and 600 released AEGIS rows. The 2Wiki manifest records exactly 7,000/7,000 expected traces; its 200 pilot IDs are disjoint from the 1,000 confirmatory IDs.

Every run has a YAML config, seed and saved query IDs; folds and resampling use query/task groups. Traces are content-addressed, result JSONs are versioned rather than overwritten, and raw out-of-fold predictions plus reproducible bootstrap/permutation seeds are retained. The AEGIS repository commit is pinned above. A machine-readable experiment registry and a separate claim freeze record the statistic, threshold and confirmatory/diagnostic/post-hoc status. An invalid source-matching rerun that inherited stale clean metrics is retained under an explicit `invalid_` filename and excluded from every figure and claim. Figures are regenerated from result JSONs; a verification script checks every number added in this revision against those files. The released suite currently collects 39 test cases across seven test modules, including 23 parametrized negative-control cases.

Table 15: Artifact registry for the analyses added or materially changed in this revision. "Primary" means the contrast and interpretation were frozen before the confirmatory run; the AEGIS mechanism contrast is retained as a diagnostic without a V2 verdict.

Analysis	Rows / groups	Unit	Status	Primary result artifact
Hotpot/GSM8K validity campaign	24,000 / 800 test queries	query	rebuilt confirmatory	`E4_validity_audit_v2.json`
Effect-matched V2	2,352 / 200 queries	query	primary + declared robustness	`matched_effect_source_v3.json`; `matched_effect_grounding.json`
Hotpot query-matched V3	71 natural groups; five synthetic seeds	query	corrected primary	`c2st_matched_v2.json`
Hotpot V4 uncertainty	seven candidates, T0–T2	query	primary bootstrap	`decision_validity_v3.json`
2Wiki V3/V4 replication	7,000 / 1,000 confirmatory queries	query	preregistered replication	`2wiki_c2st_matched_v3.json`; `2wiki_decision_validity_v3.json`
Causal restoration	90 Hotpot; 187 2Wiki interventions	query	preregistered intervention	`counterfactual_v2.json`; `2wiki_counterfactual_v3.json`
External AEGIS audit	600 / task-disjoint folds	task	primary V1; diagnostic V2	`external_aegis_audit_v3.json`
Hotpot k=3	1,400 traces	query	operating-point replication	`k3_decision_validity_v2.json`

Reproduction order. The intended sequence is (1) validate trace manifests and query-ID disjointness, (2) rebuild features from immutable traces, fitting imputation and scaling only within each training fold, (3) emit raw out-of-fold predictions before any bootstrap or permutation summary, (4) regenerate tables and figures from the versioned JSONs, and (5) run `scripts/verify_revision_numbers.py` before compiling the manuscript. Expensive trace generation is therefore separable from the deterministic analysis/verification pass. The submission bundles copy the same four vector figures and strengthened appendix; venue-specific sources change framing and page budget, not numerical evidence.

Automated failure guards. The test suite checks that all four families produce distinct tiers and vary with seed, that the hardened tool fault leaves no tool-health flag, source degradation does not pin QCM to a sentinel, no evaluator arm contains labels, missing values remain missing until fold-fitted imputation, duplicate metrics are rejected, seed replicas collapse under deduplication, and family labels cannot encode tiers. Additional tests require content-addressed artifacts, real grounding donors, held-out sample sizes for headline results, a declared label-free severity vector whose weighting reduces imbalance, clean-metric invalidation after an intervention, explicit AEGIS-field exclusions, stable task grouping, shared queries for V4, and winner/regret recomputation inside every bootstrap resample. These guards encode the two self-audit failures as regressions that future changes must not reintroduce.